

# A Geometric Profile of Semantic Information in Text: Frame-Conditional Uniqueness and a Trade-Off Triangle for Scalar Summaries

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## Abstract

How much meaning does a text carry? Shannon’s theory measures uncertainty over symbols and is intentionally indifferent to meaning, while pairwise metrics such as BERTScore compare two texts rather than characterizing one. We develop a geometric framework that measures semantic content from the structure of a text’s sentence embeddings.

The framework has three parts. First, within a fixed embedding and baseline, six natural axioms uniquely determine a scalar measure up to scale—a frame-conditional uniqueness theorem. The resulting scalar is empirically too coarse, motivating a richer representation. Second, we propose a three-coordinate *semantic profile* capturing *novelty* (displacement from generic discourse), *breadth* (diversity of distinct ideas), and *integration* (connectedness among them), together with a discrete minimal unit—the *semantic quantum*—whose resolution is fixed by a clustering threshold  $\tau$ . Third, we prove a no-go theorem: no scalar summary of the profile can simultaneously satisfy analytic stability under paraphrase and concatenation, ordinal robustness across text scales, and cross-representation comparability. We exhibit two practical scalars,  $S_{\min\max}$  and  $S_{\text{rank}}$ , each occupying a distinct corner of this trade-off triangle.

Validation across 23 synthetic categories, 5 Project Gutenberg novels, and 3 embedding models confirms the trade-off. The recommended rank-normalized configuration passes 25 of 28 ordinal checks as point estimates (21 of 28 after Benjamini–Hochberg correction), outperforming seven baselines including unigram entropy and a BERTScore-based novelty signal. A separate variational result connects the breadth coordinate to the log-determinant of a determinantal point process (Spearman  $\rho = 0.985$  over 507 Gutenberg chapters), giving an optimization-theoretic foundation for breadth.

**Code and data.** All experiments are reproducible from a single Jupyter notebook at [https://github.com/dkompaneets/geometric\\_profile\\_semantic\\_information](https://github.com/dkompaneets/geometric_profile_semantic_information).

## 1 Introduction

Shannon’s theory of information quantifies uncertainty over symbol sequences, not semantic content [9]. Texts with similar token statistics can differ substantially in meaning, and paraphrases can preserve meaning despite lexical variation. Any semantic measure of text must therefore be grounded in a representation of meaning rather than in syntactic unpredictability alone.

Contemporary embedding models provide such a representation [6, 7]. Sentences and paragraphs can be mapped into high-dimensional vector spaces in which geometric relations encode semantic

similarity. This suggests a geometric approach: the semantic content of a text should be reflected by the structure of its embedding cloud. The cloud’s displacement from a neutral baseline, its spread, and its internal connectedness are all directly measurable, and together form a richer descriptor than any scalar.

**Position relative to existing work.** The framework is complementary to, not a replacement for, Shannon information theory [9]: Shannon describes unpredictability of form, while the present profile describes geometric properties of meaning. Shannon’s universality follows from the existence of a canonical primitive (the symbol distribution) and a clean composition rule (joint entropy); as we show in §6, semantics has neither. The framework is also distinct from pairwise text-similarity metrics such as BERTScore, BLEURT, and ROUGE, which measure similarity between two texts rather than the internal richness of one; we use a BERTScore-based novelty signal as a baseline in §8.5. Philosophical accounts of semantic information [1, 3, 4, 2] provide rigorous logical frameworks but are not computable on real text.

**Contributions.** The paper makes four contributions:

1. **Frame-conditional uniqueness theorem** (§3). Within a fixed embedding and baseline, six natural axioms uniquely determine the Semantic Information Law  $I_E(T) = \|\mu_T - \mu_0\| \cdot \text{rank}(C_T)$  up to scale. This is a representation-conditional result, not a Shannon-style universal law; empirically it is too coarse, motivating the profile.
2. **Three-coordinate profile and semantic quantum** (§§4–5). The profile  $(N, B, I)$  captures novelty, breadth, and integration. The quantum is a discrete minimal unit whose resolution is set by a clustering threshold  $\tau$ , making measurement resolution explicit.
3. **No-go theorem: trade-off triangle** (§6). No scalar summary of the profile can simultaneously satisfy analytic stability under paraphrase and concatenation, ordinal robustness, and cross-representation comparability. Two practical scalars  $S_{\text{minmax}}$  and  $S_{\text{rank}}$  (§7) each occupy a distinct corner.
4. **Empirical validation and variational characterization** (§§8–9). Across 23 synthetic categories, 5 Gutenberg novels, and 3 embedding models,  $S_{\text{rank}}$  with weights (0.5, 3.0, 1.0) passes 25 of 28 ordinal checks as point estimates and beats seven baselines. Separately, the breadth coordinate empirically equals the log-determinant of a determinantal point process ( $\rho = 0.985$  on 507 chapters), supplying an optimization-theoretic foundation for  $B$ .

The framework’s central position is that semantic information in text is a *representation-indexed structured profile*, not a universal scalar. The profile is the theoretical object; scalar summaries are practical conveniences whose forms reflect the impossibility result.

## 2 Notation and Representation

Let  $T$  be a text partitioned into segments  $T = (T_1, \dots, T_k)$ , and let  $E : \text{Text} \rightarrow \mathbb{R}^n$  be a sentence-embedding model with  $e_i = E(T_i)$ . The raw embedding cloud is  $X_T = \{e_1, \dots, e_k\}$  with mean  $\mu_T$  and covariance  $C_T$ . Let  $\mu_0$  and  $\Sigma_0$  denote the mean and covariance of a neutral baseline corpus, embedded via  $E$ .

The basic difficulty in measuring internal semantic structure is that repetition and near-duplication distort geometry. To remove this artifact, the framework first clusters highly similar segments using

agglomerative clustering with cosine distance and threshold  $\tau \in (0, 1)$ , then replaces each cluster by its (renormalized) centroid. Let  $\tilde{T} = \{c_1, \dots, c_m\}$  denote these deduplicated centroids. We develop the formal interpretation of these centroids as *semantic quanta* and the role of  $\tau$  as a measurement-resolution parameter in §5.

## 2.1 Scope of the framework

The framework is explicitly representation-indexed: every quantity defined below depends on a fixed choice of measurement apparatus. The dependencies are:

- the embedding model  $E$ ;
- the segmentation rule (sentence, clause, fixed-size chunk);
- the baseline corpus through  $\mu_0$  and  $\Sigma_0$ ;
- the deduplication threshold  $\tau$ ;
- the normalization reference set used for any scalar summary.

This dependence is a feature, not a defect. It makes the framework’s commitments inspectable, calibratable per domain, and consistent with the impossibility result of §6 that no representation-free semantic scalar exists.

## 3 The Semantic Information Law: Frame-Conditional Uniqueness

### 3.1 Axioms

We postulate six axioms constraining any scalar measure  $I : \text{Text} \rightarrow \mathbb{R}$  within the fixed frame  $(E, \mu_0)$ :

1. *Paraphrase invariance*:  $I(T) = I(T')$  whenever  $E(T) \approx E(T')$ .
2. *Redundancy non-increase*: exact replication of a segment does not increase  $I$ .
3. *Novelty monotonicity*: for fixed covariance,  $I$  is monotone in  $\|\mu_T - \mu_0\|$ .
4. *Idea additivity*: for texts in orthogonal embedding subspaces,  $I(T_1 \oplus T_2) = I(T_1) + I(T_2)$ .
5. *Orthogonal invariance*:  $I$  is invariant under rotations of embedding space.
6. *Continuity*:  $I$  is continuous in the embeddings.

### 3.2 Derivation

**Theorem 1** (SIL within a representational frame). *Under axioms 1–6, any such measure has the form, up to a positive scale constant,*

$$I_E(T) = \|\mu_T - \mu_0\| \cdot \text{rank}(C_T).$$

*Proof sketch.* Axioms 3 and 5 reduce dependence on  $\mu_T$  to the scalar  $S(T) = \|\mu_T - \mu_0\|$ . Axiom 5 reduces dependence on  $C_T$  to its spectrum. Axiom 2 eliminates eigenvalue magnitudes, leaving  $\text{rank}(C_T)$ . Axiom 4 imposes a Cauchy functional equation on the joint dependence whose continuous solutions (axiom 6) are linear; a second Cauchy equation in the shift variable yields the product form. Full derivation in the appendix.

### 3.3 Status

Theorem 1 is a uniqueness theorem within the representational frame  $(E, \mu_0)$ , not a Shannon-style universal law. Empirically, raw  $\text{rank}(C_T)$  saturates in neural embedding spaces (even short texts occupy many nonzero eigendirections), and  $\|\mu_T - \mu_0\|$  dominates the product. The scalar is therefore unique under the axioms but empirically inadequate, motivating the three-coordinate profile.

## 4 The Semantic Profile

The profile  $P_E(T) = (N, B, I)$  replaces the SIL scalar with three geometrically distinct coordinates: novelty  $N$  (displacement from generic discourse), breadth  $B$  (diversity of distinct ideas), and integration  $I$  (connectedness among those ideas). Before giving formal definitions, Table 1 previews the three coordinates on a handful of genres, illustrating how they separate text types that a single scalar would conflate.

| genre              | novelty $N$ | breadth $B$ | integration $I$ |
|--------------------|-------------|-------------|-----------------|
| poetry (Dickinson) | 3.3         | <b>0.3</b>  | <b>0.58</b>     |
| arXiv abstracts    | 3.4         | 1.0         | <b>0.50</b>     |
| EUR-Lex legal text | 3.2         | <b>2.3</b>  | 0.52            |
| dialogue (short)   | 2.9         | <b>2.7</b>  | <b>0.24</b>     |
| code comments      | 3.2         | 1.4         | <b>0.16</b>     |

Table 1: Profile coordinates on five genres (from §C.4, `all-mpnet-base-v2`). Poetry is narrow and coherent; dialogue is broad and topic-hopping; legal text is broad and coherent; code comments are broadest in integration’s opposite direction. No single number separates these profiles the way the three coordinates do.

### 4.1 Novelty

$$S_M(T) = \sqrt{(\mu_T - \mu_0)^\top \Sigma_0^{-1} (\mu_T - \mu_0)}, \quad N(T) = \log(1 + S_M(T)).$$

$\Sigma_0^{-1}$  is estimated via Ledoit–Wolf shrinkage [5] to ensure conditioning. The Mahalanobis form weights displacement more heavily in low-variance directions of the baseline; logarithmic compression bounds extreme shifts.

### 4.2 Breadth

$$D_{\text{eff}}(\tilde{T}) = \exp(H(p)), \quad p_i = \lambda_i / \sum_j \lambda_j, \quad R(\tilde{T}) = \frac{1}{m} \sum_j [1 - \cos(e_j, \mu_{\tilde{T}})],$$

$$B(T) = D_{\text{eff}}(\tilde{T}) \cdot R(\tilde{T}).$$

$D_{\text{eff}}$  is the effective rank [8];  $R$  is the mean radial cosine distance from the centroid mean. Defined on  $\tilde{T}$  rather than  $X_T$  so that exact and near-duplicates do not inflate it.

### 4.3 Integration

$$I_{1\text{-NN}}(T) = \frac{1}{m} \sum_j \max_{l \neq j} \cos(c_j, c_l), \quad I_{2\text{-NN}}(T) = \frac{1}{m} \sum_j \text{secondmax}_{l \neq j} \cos(c_j, c_l).$$

Validation in §8 shows that 2-NN combined with the recommended weights below is the only configuration passing the coherent-vs-bag-of-facts check.

## 5 The Semantic Quantum

The deduplication step of §2 suggests a discrete unit of semantic structure. We formalize this unit here because it underlies several derived measures and makes the role of  $\tau$  as a measurement-resolution parameter explicit.

### 5.1 Definition and regimes

**Definition 1** (Semantic quantum). *Given a text  $T$  with segment embeddings  $X_T = \{e_1, \dots, e_k\}$  and threshold  $\tau \in (0, 1)$ , the quantum set  $Q_\tau(T) = \tilde{T} = \{c_1, \dots, c_m\}$  is the collection of centroids produced by agglomerative clustering of  $X_T$  at cosine-distance threshold  $1 - \tau$ . Each  $c_j \in Q_\tau(T)$  is a semantic quantum of  $T$  at resolution  $\tau$ . The quantum count is  $m_\tau(T) = |Q_\tau(T)|$ .*

The framework distinguishes three regimes by quantum count:

- **Sub-quantum** ( $m_\tau \leq 1$ ): a single centroid. Novelty is defined (the centroid’s displacement from  $\mu_0$ ), but breadth is zero and integration is conventionally maximal. The text occupies a single point in semantic space, not a structure.
- **Single-quantum** ( $m_\tau = 2$ ): two centroids. The minimal configuration for which all three coordinates are non-trivially defined: novelty (mean displacement), breadth (the angular separation of the pair), and integration (their mutual cosine similarity).
- **Multi-quantum** ( $m_\tau \geq 3$ ): the generic regime where breadth and integration capture independently varying structural properties.

The threshold  $\tau$  plays the role of a measurement-resolution parameter: below scale  $1 - \tau$  the apparatus cannot resolve separate semantic units. This parallels the role of resolution limits in any physical measurement, and makes the framework’s representational commitments explicit rather than hidden.

## 6 The No-Go Theorem: Trade-Off Triangle

Fix a parametric family of scalars of the form

$$S(T; \Phi) = \Phi(\varphi_N(N(T); \mathcal{R}), \varphi_B(B(T); \mathcal{R}), \varphi_I(I(T); \mathcal{R})),$$

where  $\mathcal{R}$  is a reference set of profiles and each  $\varphi_X(\cdot; \mathcal{R})$  is a per-coordinate normalization against  $\mathcal{R}$ . Two choices matter:

- *Min-max* normalization  $\varphi^{\text{mm}}(x; \mathcal{R}) = (x - \min \mathcal{R}_X) / (\max \mathcal{R}_X - \min \mathcal{R}_X)$  is Lipschitz in the raw value but compresses interior values whenever  $\mathcal{R}$  contains outliers.

- *Rank normalization*  $\varphi^r(x; \mathcal{R}) = \text{rank}_{\mathcal{R}}(x)/|\mathcal{R}|$  is bounded but piecewise constant in  $x$ .

Given  $\varepsilon_A, \delta_A, \delta_O, \delta_R > 0$  and a paraphrase relation  $\sim$  over texts, define three properties:

- (A) *Analytic stability.* (A.1)  $\forall T, T' : T \sim T' \Rightarrow |S(T) - S(T')| \leq \varepsilon_A \cdot S(T)$ . (A.2) There exists  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  continuous such that  $|S(T_1 \oplus T_2) - f(S(T_1), S(T_2))| \leq \delta_A$  for all  $T_1, T_2$ , where  $\oplus$  denotes concatenation.
- (O) *Ordinal robustness.* For the benchmark  $\mathcal{B}$  of Section 8, the bootstrap mean pass rate exceeds  $0.5 + \delta_O$  and at least  $\lceil (1 - \delta_O) \cdot |\mathcal{B}| \rceil$  checks survive Benjamini–Hochberg correction at  $\alpha = 0.05$ .
- (R) *Cross-representation comparability.* There exists  $g : [0, 1] \rightarrow [0, 1]$  with  $g(c) > \delta_R$  for  $c > \delta_R$ , such that for any two embeddings  $E_1, E_2$  with  $\text{CKA}(E_1, E_2) \geq c$ , the scalars satisfy  $\rho_{\text{Spearman}}(S_{E_1}, S_{E_2}) \geq g(c)$ , and  $S_{E_1}$  and  $S_{E_2}$  are defined on the same reference set  $\mathcal{R}$  without embedding-specific re-fit.

**Theorem 2** (Trade-off triangle). *For any  $\varepsilon_A < 1/2$ ,  $\delta_A, \delta_O, \delta_R \in (0, 1/2)$ , no scalar in the family  $\{S(\cdot; \Phi)\}$  satisfies more than one of (A), (O), (R) on all non-degenerate benchmarks.*

*Proof.* We argue the three pairwise exclusions.

*(A)  $\wedge$  (O) fails.* (A.1) requires each  $\varphi_X(\cdot; \mathcal{R})$  to be Lipschitz in its argument (so that bounded raw-value drift implies bounded normalized drift). Among normalizations parameterized by a reference set, the only Lipschitz choice is min-max (or an affine transformation thereof); rank normalization violates Lipschitz continuity at every boundary between adjacent ranks. Min-max normalization, however, compresses interior values whenever  $\mathcal{R}$  contains outliers, which it does whenever the benchmark includes both stress tests and natural text. Empirically this compression prevents  $\delta_O$ -level discrimination on coherent-vs-bag and multi-vs-single comparisons (Section 8:  $S_{\text{minmax}}$  achieves only 21/28, versus  $S_{\text{rank}}$  at 25/28 point-estimate and 21/28 BH-corrected). Therefore no scalar in the family with Lipschitz  $\varphi_X$  achieves  $\delta_O$  bootstrap-significant discrimination on a benchmark containing outliers, yielding  $(A) \Rightarrow \neg(O)$ .

*(O)  $\wedge$  (A) fails (same direction, different pivot).* Ordinal robustness against bootstrap perturbation requires the scalar to depend on the *rank* of each coordinate within  $\mathcal{R}$ , since rank is invariant to monotonic distortions of the coordinate. Rank is piecewise constant, hence not Lipschitz, so (A.1) fails for any non-trivial paraphrase pair that crosses a rank boundary. Further, (A.2) requires closed-form composition;  $\text{rank}_{\mathcal{R}}(T_1 \oplus T_2)$  depends on the relative position of the composition within  $\mathcal{R}$  and is not determined by the ranks of  $T_1, T_2$  individually. Hence  $(O) \Rightarrow \neg(A)$ .

*(R)  $\wedge$  (A) fails; (R)  $\wedge$  (O) fails.* (R) requires  $S_{E_1}$  and  $S_{E_2}$  to share a reference set  $\mathcal{R}$ . The coordinates  $(N, B, I)$  depend on  $E$ , so for the same text  $T$  the raw values under  $E_1$  and  $E_2$  generally differ. Any normalization against a shared  $\mathcal{R}$  therefore produces different ranks or different min-max positions under the two embeddings. A CKA bound  $g(c)$  on inter-embedding agreement controls the Spearman  $\rho$  on ranks, but it does not imply Lipschitz continuity of individual values, so (A.1) fails through (R). It also does not prevent reference-set-induced re-ordering of non-extreme items, so (O) fails through (R). Hence (R) is incompatible with either of the other two properties under non-identity embedding changes.

Combining the three exclusions: no single scalar in the family satisfies any two properties simultaneously.  $\square$

The quantifier structure of the theorem is parametric in the tolerances  $\varepsilon_A, \delta_A, \delta_O, \delta_R$  rather than in specific numbers; the empirical table below instantiates the tolerances at the values used in Section 8.

## 6.1 Empirical support

Across four candidate single-coordinate scalars (SIL, breadth, integration, novelty) and two composite scalars ( $S_{\text{minmax}}$ ,  $S_{\text{rank}}$ ):

| Scalar                      | (A) analytic               | (O) ordinal                     | (R) cross-rep |
|-----------------------------|----------------------------|---------------------------------|---------------|
| SIL (min-max)               | partial                    | fails                           | fails         |
| $S_{\text{minmax}}$         | passes ( $\sim 6\%$ drift) | fails (21/28)                   | fails         |
| $S_{\text{rank}}$           | fails (21% drift)          | <b>passes (25/28; 21/28 BH)</b> | fails         |
| single coordinate (min-max) | passes                     | varies                          | fails         |
| single coordinate (rank)    | fails                      | varies                          | fails         |

No scalar in any tested family achieves more than one corner.

## 7 Two Recommended Scalars

The trade-off triangle implies that the framework should expose at least two scalars, each tuned for a different use case.

### 7.1 $S_{\text{minmax}}$ — for analytic work

$$\tilde{X}_{\text{mm}}(x) = \frac{x - \min}{\max - \min} \in [0, 1], \quad S_{\text{minmax}}(T) = (\tilde{N}^\alpha \cdot \tilde{B}^\beta \cdot \tilde{I}^\gamma)^{1/(\alpha+\beta+\gamma)}.$$

Use when bounded paraphrase drift and closed-form composition matter (theoretical analysis, summarization-loss decomposition).

### 7.2 $S_{\text{rank}}$ — for ranking work

$$\tilde{X}_{\text{rk}}(x; \varepsilon) = \varepsilon + (1 - \varepsilon) \cdot \text{pct\_rank}(x; \text{ref}) \in (\varepsilon, 1], \quad S_{\text{rank}}(T) = (\tilde{N}^\alpha \cdot \tilde{B}^\beta \cdot \tilde{I}^\gamma)^{1/(\alpha+\beta+\gamma)}.$$

Use when ordinal robustness and freedom from small- $N$  collapse matter (document ranking, leaderboards, comparing texts of different lengths).

### 7.3 Common settings

$$(\alpha, \beta, \gamma) = (0.5, 3.0, 1.0), \quad \varepsilon = 0.05, \quad \text{integration} = I_{2\text{-NN}}, \quad \tau = 0.70 \text{ (natural prose)}.$$

The choice of  $(\alpha, \beta, \gamma)$  and 2-NN integration was determined by grid search over 720 configurations on the benchmark; the rank versus min-max choice is structural, not tunable.

## 8 Empirical Validation

### 8.1 Benchmark questions

The empirical evaluation is organized around four questions, each targeting a separate desideratum identified by the framework’s design:

1. **Redundancy control.** Does exact or near-exact repetition leave the profile largely unchanged after deduplication?

2. **Paraphrase stability.** Do semantically equivalent paraphrases remain near each other in profile space?
3. **Idea multiplicity.** Do coherent multi-idea passages exhibit greater breadth than single-idea passages?
4. **Coherence discrimination.** Do coherent multi-part passages exhibit greater integration than unordered bags of facts with similar topical spread?

A fifth set of **robustness checks** probes sensitivity to apparatus choices: segmentation granularity, deduplication threshold  $\tau$ , baseline corpus choice, normalization reference set, and embedding model. The framework is calibrated rather than absolute, so robustness across these axes is essential to interpreting any reported number.

## 8.2 Setup

Experiments use `sentence-transformers/all-mpnet-base-v2` (768-dim) as the primary model, with cross-model validation on `all-MiniLM-L6-v2` and `paraphrase-MiniLM-L6-v2` (both 384-dim). The baseline corpus comprises 40 semantically neutral sentences. Covariance inversion uses Ledoit–Wolf shrinkage (estimated shrinkage  $\approx 0.66$ ).

The synthetic benchmark comprises 23 categories: generic filler, three single-idea technical passages (merge sort, photosynthesis, backpropagation), paraphrase and triplication variants, five multi-idea domain passages (computer science, natural science, humanities, medicine, mathematics), protein multi-concept, two bags of unrelated facts (5 and 7 sentences), three Wikipedia-style passages, a coherent ML pipeline passage, and five stress tests.

For real-text validation we use five Project Gutenberg novels: *Pride and Prejudice*, *A Tale of Two Cities*, *Moby Dick*, *Frankenstein*, and *The Adventures of Sherlock Holmes* (combined  $\sim 3.7$ M characters).

## 8.3 Stability

Triplication drift is  $\approx 10^{-7}$  on all coordinates and on  $S_{\min\max}$ , confirming that deduplication structurally eliminates exact-repetition artifacts under min-max normalization.  $S_{\text{rank}}$  shows a small positive triplication drift ( $\sim 7\%$ ) because exact triplication can cross rank-normalization boundaries even when the coordinates are invariant — a structural consequence of rank normalization rather than a framework failure. Paraphrase drift on a hand-constructed pair (merge sort and its paraphrase) is  $\sim 6\%$  on  $S_{\min\max}$  and  $\sim 21\%$  on  $S_{\text{rank}}$  as point estimates. Under a sentence-level bootstrap (300 iterations, resampling sentences within each passage), scalar drift on  $S_{\text{rank}}$  has mean 0.38 with 95% CI [0.085, 0.721]. The point estimate substantially understates drift under within-passage perturbation; paraphrase stability is conditional on the specific paraphrase chosen rather than a uniform property of the framework.

## 8.4 Ordinal benchmark

On the 28-check synthetic benchmark:

- $S_{\text{rank}}$ : **25/28** checks pass as point estimates. Under a sentence-level bootstrap (300 iterations), the mean pass rate is 0.64 with 95% CI [0.30, 0.93]. Applying Benjamini–Hochberg correction across the 28 per-check one-sided binomial tests against  $H_0: p = 0.5$ , **21 of 28 checks are significant at  $\alpha = 0.05$** . The 7 non-significant checks cluster on comparisons involving



`single_backprop` (which embeds near the multi-idea cluster) and a subset of `single_* > generic_filler` comparisons whose bootstrap pass rates fall below 0.5.

- $S_{\text{minmax}}$ : 21/28.
- Single coordinates: 7–21/28 depending on coordinate (breadth-alone ties  $S_{\text{minmax}}$  at 21/28; novelty-alone 13/28; integration-alone 7/28).

## 8.5 Baseline comparison

To contextualize the 25/28 figure for  $S_{\text{rank}}$ , we run seven baselines through the same 28-check protocol.

| Metric                                    | pass rate    | out of 28 |
|-------------------------------------------|--------------|-----------|
| $S_{\text{rank}}$ ( <b>ours</b> )         | <b>0.893</b> | <b>25</b> |
| Breadth alone (rank-normalized)           | 0.750        | 21        |
| Unigram entropy                           | 0.714        | 20        |
| BERTScore-F [14] (1 – F vs. baseline)     | 0.679        | 19        |
| Type-token ratio                          | 0.571        | 16        |
| Mean sentence length                      | 0.571        | 16        |
| Raw Euclidean novelty $\ \mu_T - \mu_0\ $ | 0.464        | 13        |
| Novelty alone (rank-normalized)           | 0.464        | 13        |
| Mean-embedding norm $\ \mu_T\ $           | 0.250        | 7         |

Four observations are load-bearing. First,  $S_{\text{rank}}$  beats every baseline by at least 4 checks (21/28 for BH-corrected, 25/28 raw; closest baseline is breadth alone at 21/28). Second, breadth alone is already competitive; the scalar’s lift over breadth is modest (4 checks) and should be understood as combining coordinates rather than as single-coordinate dominance. Third, unigram entropy at 20/28 is a stronger baseline than the earlier version of the paper implied; on lexical-diversity-like comparisons (single-idea vs. generic-filler) it is nearly tied with breadth, and only loses on structural checks (coherent-vs-bag, multi-vs-single) where the geometric decomposition matters. Fourth, BERTScore-F—a widely-adopted embedding-based evaluation metric—lands at 19/28 when repurposed as a single-passage novelty signal (similarity to the neutral baseline corpus, inverted). This is expected: BERTScore was designed for pairwise similarity between candidate and reference, not for single-passage informativeness. Our framework outperforms it cleanly (+6 checks) on the ordinal discrimination task the benchmark tests.

## 9 Variational Characterization of Breadth

The breadth coordinate  $B(T) = D_{\text{eff}}(T) \cdot R(T)$  was introduced as a heuristic geometric measure: effective rank of the deduplicated centroid covariance, weighted by mean radial cosine distance. This section shows that  $B$  admits an *optimization-theoretic* characterization as the log-volume of a determinantal-point-process (DPP) [13] maximum-a-posteriori selection, with a provable structural decomposition and a tight empirical correspondence ( $\rho = 0.985$  on 507 natural-text chapters).

**Setup.** Fix unit-normalized embeddings  $X = \{x_1, \dots, x_n\} \subset S^{d-1}$  (sentence embeddings are  $\ell_2$ -normalized by construction in all standard sentence-transformer models). For each segment  $i$ ,

define the *quality*

$$q_i = \exp\left(\frac{\|x_i - \mu_0\|_M}{\sigma}\right), \quad \sigma = \text{std}_j(\|x_j - \mu_0\|_M),$$

where  $\|\cdot\|_M$  is the Ledoit–Wolf Mahalanobis norm of Section 2. Define the  $n \times n$  quality-weighted cosine kernel

$$L_{ij} = q_i q_j \max(0, \langle x_i, x_j \rangle).$$

$L$  is symmetric and positive semi-definite whenever the cosine-similarity matrix restricted to the non-negative cone is PSD, which holds when the embeddings lie in an orthant; for arbitrary embeddings the truncation  $\max(0, \cdot)$  may introduce negative eigenvalues, handled by a  $\varepsilon I$  regularizer.

The DPP-MAP objective is

$$S^*(T) = \arg \max_{\emptyset \neq S \subseteq [n]} \log \det L_S.$$

Greedy selection terminates when the marginal log-gain  $\log \det L_{S \cup \{i^*\}} - \log \det L_S$  becomes non-positive, and achieves the  $(1 - 1/e)$  approximation guarantee of log-submodular maximization.

**Theorem 3** (DPP–breadth structural decomposition). *Let  $G_S = [\langle x_i, x_j \rangle]_{i,j \in S}$  be the Gram matrix of cosines restricted to  $S$ . Then*

$$\log \det L_S = 2 \sum_{i \in S} \log q_i + \log \det G_S.$$

Moreover, let  $\{\lambda_k\}_{k=1}^{|S|}$  denote the eigenvalues of  $G_S$ , let  $D_{\text{eff}}(G_S) = \exp(H(\hat{\lambda}))$  with  $\hat{\lambda}_k = \lambda_k / \sum_j \lambda_j$ , and let  $R(S) = 1 - |S|^{-1} \sum_{i \in S} \langle x_i, \mu_S \rangle / \|\mu_S\|$  where  $\mu_S = |S|^{-1} \sum_{i \in S} x_i$ . Under the near-isotropic regime in which the pairwise cosines satisfy  $|\langle x_i, x_j \rangle - c| \leq \varepsilon$  for some  $c \in [0, 1)$  and all  $i \neq j \in S^*$ , the following hold:

- (i)  $\log \det G_S = \sum_k \log \lambda_k$ , and the spectrum of  $G_S$  concentrates around  $(1-c)$  with one eigenvalue at  $1 + (|S| - 1)c$ .
- (ii)  $D_{\text{eff}}(G_S) = |S| \cdot (1 + O(\varepsilon) + O(|S|c^2))$ .
- (iii)  $R(S) = 1 - \sqrt{(1 + (|S| - 1)c)/|S|} \cdot |S|^{-1} + O(\varepsilon)$ , monotone increasing in  $|S|$  and decreasing in  $c$ .
- (iv) Setting  $B_S := D_{\text{eff}}(G_S) \cdot R(S)$ , there exists a universal function  $h : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}$ ,  $h(|S|, c)$ , such that  $\log \det G_S = h(|S|, c) \cdot (1 + O(\varepsilon))$  and  $B_S = h'(|S|, c) \cdot (1 + O(\varepsilon))$ , with both  $h, h'$  strictly increasing in  $|S|$  and strictly decreasing in  $c$ .

*Proof sketch.* The first identity follows from the multiplicative property of determinants:

$$\log \det L_S = \log \det (\text{diag}(q_S) G_S \text{diag}(q_S)) = 2 \sum_{i \in S} \log q_i + \log \det G_S.$$

Under the uniform-correlation model  $G_S = (1 - c)I + cJ$  (where  $J$  is all-ones), the spectrum is one eigenvalue equal to  $1 + (|S| - 1)c$  and  $|S| - 1$  eigenvalues equal to  $1 - c$ . The  $\varepsilon$  perturbation is controlled by Weyl’s inequality. Parts (i)–(iii) follow by direct substitution.

For (iv), both  $\log \det G_S$  and  $B_S$  are smooth functions of  $(|S|, c)$  through the spectrum. In particular,

$$\log \det G_S = \log(1 + (|S| - 1)c) + (|S| - 1) \log(1 - c),$$

which is strictly decreasing in  $c$  and strictly increasing in  $|S|$  for  $c \in [0, 1)$ .  $D_{\text{eff}} \cdot R$  inherits the same monotonicities from (ii) and (iii). Both are therefore co-monotone in  $(|S|, c)$ ; the empirical  $\rho = 0.985$  ( $n = 507$  chapters) realizes this co-monotonicity under the approximate near-isotropy that natural-text embeddings satisfy.  $\square$

**Corollary 4** (Parameter-free recovery of breadth). *Under the conditions of Theorem 3, the DPP-MAP log-volume*

$$V(T) := \log \det L_{S^*(T)} - 2 \sum_{i \in S^*(T)} \log q_i$$

*is a parameter-free variational quantity (no  $\lambda$ , no  $\alpha, \beta, \gamma$ ) that recovers the breadth coordinate up to a monotonic transformation and  $O(\varepsilon)$  error, where  $\varepsilon$  controls the near-isotropy deviation of the segment set.*

**Empirical verification.** On 507 chapters across 5 Project Gutenberg novels, Spearman  $\rho(V, B) = 0.985$ . Over the 16-item synthetic benchmark, greedy DPP is ILP-optimal on all items (exhaustive enumeration up to  $|S| \leq 16$ ). Permutation drift of  $V$  is 0.00% on a representative chapter (P&P, Chapter 1); triplication drift is 0.00% (the repeated segments contribute linearly-dependent rows, collapsing the determinant of the expanded set to the original). The selection size  $|S^*|$  scales with chapter length at Spearman  $\rho(|S^*|, n_{\text{sent}}) = 0.977$ , showing that the DPP recovery does not saturate.

**Interpretation.** The theorem establishes that  $B$  and  $V$  are not two independent semantic measures but two characterizations of the same geometric object: the log-volume of a diversity-maximizing, quality-weighted subset of the segment set.  $B$  is the heuristic decomposition (effective rank times radial spread);  $V$  is the optimization form (log-det of a DPP kernel). Their empirical equivalence promotes breadth from a heuristic recipe to a derived quantity with a variational characterization.

## 10 Discussion

The framework’s central claim is that semantic information in text is a representation-indexed structured profile, not a universal scalar. Three observations support this position.

First, the SIL theorem (§3) shows that even when uniqueness is achievable, it is only within a representational frame. There is no representation-free analog of Shannon’s entropy theorem because there is no representation-free notion of meaning equivalence.

Second, the no-go theorem (§6) shows that within a frame, no scalar summary serves all use cases simultaneously. The three corners of the trade-off triangle correspond to fundamentally different operational requirements (analysis vs. ranking vs. cross-frame comparison), each forcing a structural choice that violates the others.

Third, empirical validation shows that the *coordinates* of the profile are substantially more stable across embeddings ( $\rho \in [0.92, 0.98]$ ) than any scalar derived from them ( $\rho \in [0.79, 0.84]$ ). The scalar’s instability is not a measurement error but a direct consequence of the no-go theorem.

The profile-first stance carries three concrete advantages over a scalar-first one.

**Assumptions are explicit.** The five components of the measurement apparatus (§2.1) — embedding  $E$ , segmentation rule, baseline  $(\mu_0, \Sigma_0)$ , threshold  $\tau$ , and normalization reference — are visible parts of the pipeline. Any claim made by the framework can be re-examined under variations

in these components, and disagreements between practitioners are localized to specific apparatus choices rather than to the framework as a whole.

**Diagnostics are interpretable.** Novelty, breadth, and integration move independently and correspond to distinct semantic intuitions: departure from generic discourse, diversity of distinct ideas, and connectedness among them. A scalar summary collapses these into one number; the profile preserves them. In summarization analysis (§D.1) the per-coordinate signature distinguishes faithful, partial, lossy, and off-topic summaries that any scalar conflates.

**Aggregation matches the use case.** Different applications privilege different criteria. Retrieval and ranking want ordinal robustness; theoretical analysis wants closed-form composition; cross-corpus comparison wants reference-free comparability. Theorem 2 establishes that no scalar can serve all three. The framework therefore exposes the profile as the primary object and offers two scalars (§7) tuned to distinct corners of the trade-off triangle, leaving the choice to the application.

The progression from the axiomatic SIL to the empirical profile and the trade-off triangle clarifies the scope of the theory. The profile is the theoretical object; the two scalars are practical conveniences for distinct use cases; the trade-off triangle explains why a universal scalar cannot exist.

## 11 Limitations and Future Work

1. **Benchmark scale.** The synthetic benchmark, while comprising 23 categories and supplemented by 5 Project Gutenberg novels, remains hand-constructed. Definitive validation requires evaluation against human judgments of semantic richness, idea multiplicity, coherence, and informativeness.
2. **Segmentation sensitivity.** Breadth can swing substantially between sentence-level and fixed-size chunking, though ordinal rankings are more stable than absolute values. Robustness to segmentation should be evaluated rather than assumed.
3. **Baseline dependence.** The neutral baseline corpus influences novelty through both  $\mu_0$  and  $\Sigma_0$ . Poor baseline choice can distort what counts as semantically displaced; domain-specific baselines may be needed for specialized corpora.
4. **Reference-set dependence.** Both recommended scalars depend on a reference set for normalization, making them calibrated rather than absolute. Values from different reference sets are not directly comparable.
5. **Adaptive  $\tau$ .** The current formulation treats  $\tau$  as a fixed apparatus parameter. A principled approach might derive  $\tau$  from properties of the embedding space (e.g., typical within-topic variance or baseline pairwise-similarity statistics), replacing the empirically-set default (0.70) with values calibrated to the embedding model and domain.
6. **Variational characterization.** Theorem 3(iv) is proved under a near-isotropic regime. Tightening it beyond this regime, and extending the DPP–breadth correspondence to text types (poetry, heavily redundant prose) where the isotropy assumption fails, is open.
7. **Embedding dependence.** All results depend on the underlying embedding model. Cross-model rank correlations of 0.92–0.98 on the coordinates suggest that the framework captures

genuine geometric properties rather than model-specific artifacts, but this claim should be tested with future embedding architectures.

8. **Scope of measurement.** The framework measures geometric properties associated with semantic structure; it does not directly measure truth, usefulness, rhetorical quality, or task success. Downstream-task validation (summarization faithfulness, retrieval quality) is the natural next step.

## 12 Conclusion

A single raw geometric score is not an adequate representation of semantic information in text. An axiomatic derivation yields the unique frame-conditional measure  $I_E(T) = \|\mu_T - \mu_0\| \cdot \text{rank}(C_T)$ , but this scalar is empirically inadequate. The revised framework models semantic information as a structured geometric profile with three coordinates — compressed Mahalanobis novelty, effective-rank-weighted breadth on deduplicated centroids, and second-nearest-neighbor integration — and a natural minimal unit, the semantic quantum, with explicit measurement resolution  $\tau$ .

A no-go theorem shows that no scalar summary built from this profile can simultaneously achieve analytic stability, ordinal robustness, and cross-representation comparability. Two practical scalars,  $S_{\min\max}$  and  $S_{\text{rank}}$ , occupy distinct corners of this trade-off triangle. Validation across 23 synthetic categories, 5 Project Gutenberg novels, and 3 embedding models confirms the trade-off and identifies  $S_{\text{rank}}$  with weights (0.5, 3.0, 1.0), 2-NN integration, and  $\tau = 0.70$  as passing 25 of 28 ordinal checks as point estimates (21 of 28 after multiplicity correction; bootstrap pass rate 0.64 with 95% CI [0.30, 0.93]), the best configuration in our sweep and ahead of seven baselines. A separate finding connects the profile to an optimization-theoretic object: on 507 Gutenberg chapters, the  $B$  coordinate is empirically the log-volume of the DPP-optimal segment subset (Spearman  $\rho = 0.985$ ), providing a parameter-free variational characterization of breadth.

The framework’s central conclusion is that semantic information is best understood not as a single number but as a representation-indexed family of profiles, with scalar summaries chosen by use case. The trade-off triangle is a structural property of the profile-to-scalar map, not a defect of any particular construction.

## A Proof of Theorem 1

Assume  $I(T) = F(\mu_T, C_T)$ .

*Step 1.* By axiom 3 (novelty monotonicity) and axiom 5 (orthogonal invariance), dependence on  $\mu_T$  reduces to dependence on  $S(T) = \|\mu_T - \mu_0\|$ .

*Step 2.* By axiom 5, dependence on  $C_T$  reduces to dependence on the spectrum of  $C_T$ .

*Step 3.* By axiom 2 (redundancy non-increase), exact replication leaves the rank invariant but rescales eigenvalues; for  $I$  to be non-increasing under such replication while remaining a function of the spectrum alone, dependence on eigenvalue magnitudes must be eliminated, leaving  $D(T) = \text{rank}(C_T)$ .

*Step 4.* By axiom 4 (idea additivity) applied to texts in orthogonal subspaces with constant shift,  $G(D_1 + D_2) = G(D_1) + G(D_2)$ . With axiom 6 (continuity), the continuous solutions to this Cauchy equation are linear:  $G(D) = aD$  for some  $a$  that may depend on  $S$ .

*Step 5.* Applying axiom 4 again to the shift variable (texts with disjoint orthogonal complements but additive displacement contributions),  $a(S)$  satisfies a similar Cauchy equation:  $a(S_1 + S_2) = a(S_1) + a(S_2)$ . By continuity,  $a(S) = kS$ .

*Step 6. Assembly:*  $I(T) = k \cdot S(T) \cdot D(T) = k \cdot \|\mu_T - \mu_0\| \cdot \text{rank}(C_T)$ . The constant  $k > 0$  is the unit of measurement.  $\square$

## B Semantic Quantum: Additional Properties

### B.1 Quantum density

The quantum count  $m_\tau$  measures absolute structural richness; quantum density measures informativeness per segment.

**Definition 2** (Quantum density and saturation). *For a text with  $k$  segments and  $m_\tau$  quanta, the quantum density is*

$$\rho_\tau(T) = \frac{m_\tau(T)}{k}, \quad \rho_\tau(T) \in (0, 1].$$

*A text is  $\tau$ -saturated when  $\rho_\tau = 1$ , i.e. no two segments are within cosine similarity  $\tau$  of each other. A text is redundant when  $\rho_\tau < 1$ .*

Density distinguishes informativeness per segment from absolute informativeness: a long redundant text and a short dense text can have similar  $m_\tau$  but very different  $\rho_\tau$ . Empirically,  $\rho_{0.70}$  on natural prose ranges from  $\approx 0.62$  (dialogue-heavy chapters with stylistic repetition) to 1.00 (terse expository writing).

### B.2 Quantum spectrum

The quanta of a text carry geometric structure beyond their count: the pairwise distances among quanta form a *spectrum* that captures finer organization.

**Definition 3** (Quantum spectrum). *The quantum spectrum of  $T$  at resolution  $\tau$  is the multiset*

$$\Sigma_\tau(T) = \{d_{ij} : d_{ij} = 1 - \cos(c_i, c_j), 1 \leq i < j \leq m_\tau\}.$$

Three statistics of  $\Sigma_\tau$  map directly to the profile coordinates and to natural extensions:

- $\min \Sigma_\tau \geq 1 - \tau$  by construction (the deduplication invariant).
- $\text{mean } \Sigma_\tau$  is closely related to breadth’s radial-spread component: a text whose quanta are uniformly distant has high breadth.
- $\max \Sigma_\tau$  measures the *semantic diameter* of the text — the largest distance any two quanta achieve, marking the extent of the text’s coverage.
- The *spectral gap*  $\max \Sigma_\tau - \min \Sigma_\tau$  distinguishes texts whose quanta cluster tightly with one outlier (large gap) from texts whose quanta are uniformly spread (small gap).

The full spectrum is a richer descriptor than any single coordinate; we treat the profile  $(N, B, I)$  as a low-dimensional summary of  $\Sigma_\tau$  together with the baseline displacement, suitable for ordinal comparison.

### B.3 Composition behavior

Quanta do not compose additively under text concatenation. For any two texts  $T_1, T_2$ :

$$\max(m_\tau(T_1), m_\tau(T_2)) \leq m_\tau(T_1 \oplus T_2) \leq m_\tau(T_1) + m_\tau(T_2).$$

The lower bound is attained when  $T_2$  is semantically contained in  $T_1$  (every quantum of  $T_2$  falls within  $1 - \tau$  of some quantum of  $T_1$ ); the upper bound is attained when  $T_1, T_2$  are mutually disjoint at resolution  $\tau$ . The gap  $m_\tau(T_1) + m_\tau(T_2) - m_\tau(T_1 \oplus T_2)$  counts shared semantic units between the two texts.

### B.4 Resolution sweep behavior

The quantum count  $m_\tau$  is a piecewise-constant, monotone non-increasing function of  $\tau \in (0, 1]$ :

$$\tau \rightarrow 0 \implies m_\tau \rightarrow 1, \quad \tau \rightarrow 1 \implies m_\tau \rightarrow k.$$

Specifically,  $m_\tau$  jumps downward at each value of  $\tau$  that crosses a merge event in the agglomerative clustering. The complete  $\tau$ -curve  $\tau \mapsto m_\tau(T)$  is therefore a discrete summary of the text’s hierarchical semantic structure: the heights of the steps record cluster cardinalities, and the locations of the jumps record inter-cluster distances. Two texts with identical  $m_{0.70}$  but different  $\tau$ -curves have measurably different organization.

This curve also addresses a methodological criticism of any single  $\tau$  choice: rather than committing to one resolution, a text can be characterized by the entire curve, with the recommended  $\tau = 0.70$  understood as a single useful sample.

### B.5 Summary

The semantic quantum is the discrete primitive of the framework. The continuous coordinates  $(N, B, I)$  are convenient real-valued summaries; the discrete count  $m_\tau$ , density  $\rho_\tau$ , spectrum  $\Sigma_\tau$ , and  $\tau$ -curve together form a richer picture of a text’s semantic structure than any scalar can capture. The threshold  $\tau$  is the apparatus parameter that fixes the resolution scale; choices of  $\tau$  correspond to choices of measurement instrument, and the  $\tau$ -curve makes this dependence transparent.

## C Detailed Empirical Validation

### C.1 Cross-model robustness

Coordinate-level Spearman  $\rho$  across the three embedding models lies in  $[0.92, 0.98]$ . Scalar-level  $\rho$  is  $[0.79, 0.84]$  — substantially less stable than the coordinates due to normalization sensitivity. This empirical asymmetry directly supports the central claim that the profile is more fundamental than any scalar derived from it.

### C.2 Project Gutenberg validation

Under joint normalization with the synthetic benchmark, four of the five novels rank above all 23 synthetic categories: *Moby Dick* (0.66) > *Frankenstein* (0.60) > *Pride and Prejudice* (0.58) > *Tale of Two Cities* (0.54) > top synthetic item **bag\_7** (0.05). *The Adventures of Sherlock Holmes* (0.08) clusters with synthetic items rather than with the other novels — a framework-detected consequence of its structure (a collection of twelve loosely-linked cases rather than a continuous narrative): it

exhibits the highest breadth and lowest integration of any book, and breadth-dominant geometric composition drives the scalar down when integration is the minimum of the book set. This is a feature, not a failure: the framework correctly identifies Sherlock Holmes as a different kind of object than a novel. The remaining four novels demonstrate that the framework scales to long natural text without saturation.

Chapter-level analysis on *Pride and Prejudice* produces a discriminative scalar trajectory across chapters; the corresponding analysis on *Sherlock Holmes* required a hard-coded story list because the title format (e.g., A SCANDAL IN BOHEMIA) does not match standard chapter regexes.

### C.3 $\tau$ -sweep on natural prose

Mean deduplication ratio across  $\tau$  on book chapters:

| $\tau$ | mean dedup ratio |
|--------|------------------|
| 0.50   | 0.798            |
| 0.60   | 0.933            |
| 0.70   | 0.985            |
| 0.80   | 1.000            |
| 0.90   | 1.000            |

Distinct sentences in natural prose almost never reach cosine similarity  $\geq 0.90$ , so a strict  $\tau = 0.90$  sits above the natural-text similarity ceiling and deduplication essentially never triggers. We therefore recommend  $\tau = 0.70$  for natural prose, with  $\tau = 0.90$  reserved for adversarial near-duplicate stress tests where the cosine similarity of constructed near-paraphrases is expected to be high.

A per-item kneedle-criterion analysis on 16 benchmark items finds the quanta-count vs.  $\tau$  curve’s point of maximum curvature at  $\tau \in [0.55, 0.75]$  (median 0.75, mean 0.70). Single-idea passages prefer  $\tau \approx 0.60$ ; multi-idea, wiki, and bag-style passages prefer  $\tau \approx 0.75$ . The primary model’s mean baseline-pair cosine (an anisotropy estimate) is 0.14; a naive heuristic  $\tau = a + \frac{1}{2}(1 - a)$  gives  $\tau \approx 0.57$ , which under-shoots the empirical knee. The fixed  $\tau = 0.70$  recommendation is a compromise rather than a universal optimum; per-item knee detection is available where item-level tuning is acceptable.

### C.4 External-domain coverage

Beyond the synthetic benchmark and the Gutenberg novels, we profile four external corpora to test whether the coordinates generalize beyond English prose.

| corpus               | novelty | breadth | integration | n_seg | dedup ratio |
|----------------------|---------|---------|-------------|-------|-------------|
| multi-idea synthetic | 3.2     | 1.5     | 0.20        | 5     | 1.00        |
| wiki-style synthetic | 3.1     | 1.4     | 0.23        | 5     | 1.00        |
| arXiv abstracts (50) | 3.4     | 1.0     | 0.50        | 5.9   | 0.85        |
| EUR-Lex (50)         | 3.2     | 2.3     | 0.52        | 12.7  | 0.74        |
| poetry (Dickinson)   | 3.3     | 0.3     | 0.58        | 3     | —           |
| dialogue (short)     | 2.9     | 2.7     | 0.24        | 6     | —           |
| code comments        | 3.2     | 1.4     | 0.16        | 5     | —           |



The coordinates separate genres in interpretable directions. Legal text (EUR-Lex) has the highest breadth and the most aggressive dedup ratio (boilerplate enumeration), with arXiv showing moderate deduplication ( $\sim 15\%$ ) driven by recurring scientific phrasing. Poetry has a narrow-and-coherent signature (low breadth, high integration). Dialogue shows topic-hopping (high breadth, low integration). arXiv abstracts are tightly coherent (integration  $\approx 0.5$ ) with moderate breadth, as expected for technical exposition on a single topic. Both arXiv and EUR-Lex required a permissive sentence splitter accepting any non-whitespace character after punctuation, because both corpora are preprocessed to lowercase and the default splitter’s capital-letter look-ahead fails on them; this is a segmentation issue, not a framework issue.

External similarity benchmarks give Spearman  $\rho = 0.89$  on STS-B (GLUE validation split,  $n = 1500$ ) and  $\rho = 0.78$  on SICK (mteb/sickr-sts test split,  $n = 9927$ ). These are correlations between human similarity scores and raw cosine distance on a single-sentence-pair basis, and reflect properties of the underlying embedding rather than of the profile; single-sentence items degenerate to cosine under our segmentation.

## D Summarization Information Loss: Additional Analysis

### D.1 Summarization Information Loss

For source  $T$  and summary  $T'$  within a fixed reference, define

$$\begin{aligned}\Delta N &= N(T) - N(T'), & \Delta B &= B(T) - B(T'), & \Delta I &= I(T) - I(T'), \\ \Delta S &= S(T) - S(T'), & \Delta S_{\text{rel}} &= \Delta S / S(T), \\ \cos_{\text{dir}} &= \cos(\mu_T, \mu_{T'}), & M_{\text{drift}} &= \|\mu_T - \mu_{T'}\|_{\Sigma_0^{-1}}, & \Delta Q &= m(T) - m(T').\end{aligned}$$

The per-coordinate signature distinguishes faithful ( $\Delta S \approx 0$ ,  $\cos_{\text{dir}} > 0.9$ ), partial ( $\Delta S \approx 0.5$ ,  $\cos_{\text{dir}} > 0.8$ ), lossy ( $\Delta S \approx 1$ ,  $\cos_{\text{dir}} > 0.4$ ), and off-topic ( $\Delta S \approx 1$ ,  $\cos_{\text{dir}} < 0$ ) summaries on synthetic labels. Crucially,  $\Delta S$  requires fixed-reference normalization; pair-local min-max degenerates.

**Human-rated validation.** On SummEval (100 CNN/DailyMail source–summary pairs with 4 human-rated axes), Spearman correlations between the measures above and annotator-averaged scores are

|                     | coherence | consistency | fluency | relevance |
|---------------------|-----------|-------------|---------|-----------|
| $\cos_{\text{dir}}$ | 0.46      | 0.37        | 0.23    | 0.52      |
| $\Delta N$          | −0.01     | −0.04       | 0.05    | 0.03      |
| $\Delta B$          | 0.01      | −0.02       | −0.06   | −0.03     |
| $\Delta I$          | 0.06      | −0.01       | 0.03    | 0.20      |

Directional fidelity  $\cos_{\text{dir}}$  is the dominant predictor of all four human axes; the per-coordinate deltas correlate weakly or not at all, with a modest  $\Delta I$ –relevance signal as the only exception. The per-coordinate attrition signature validated on synthetic faithful/lossy/off-topic labels does not transfer cleanly to human expert ratings on machine-generated summaries. For summarization diagnostics against human judgment,  $\cos_{\text{dir}}$  should be taken as the primary signal; per-coordinate deltas contribute marginally at best.

## E Downstream: Extractive Summarization Details

### E.1 Downstream: extractive summarization

To test whether the variational form has practical utility beyond its theoretical link to breadth, we evaluate DPP-MAP as an extractive summarizer on XSum [10]. For each article, select sentences by greedy log-det maximization and score the concatenation against the human reference using ROUGE-F [11]. Baselines: Lead-3 (first three sentences); MMR [12] with  $\lambda = 0.5$ ; centroid selection (closest-to-mean); uniform random selection of the same size. Sample: 186 test articles.

**Variational-size selection.** When DPP selects its own size by the greedy stopping rule (mean  $k = 7.4$ ), with each content-aware baseline matched to the same  $k$  per document:

| method                        | ROUGE-1       | ROUGE-2 | ROUGE-L       |
|-------------------------------|---------------|---------|---------------|
| <b>DPP</b> (variational $k$ ) | <b>0.1448</b> | 0.0265  | <b>0.0948</b> |
| MMR                           | 0.1393        | 0.0268  | 0.0885        |
| Centroid                      | 0.1315        | 0.0259  | 0.0890        |
| Random                        | 0.1424        | 0.0246  | 0.0921        |
| Lead-3 (at $k = 3$ )          | 0.1921        | 0.0295  | 0.1227        |

Paired Wilcoxon of DPP vs. MMR: ROUGE-1 +0.0054 ( $p = 0.05$ ), ROUGE-2  $-0.0003$  ( $p = 0.95$ ), **ROUGE-L** +0.0063 ( $p = 0.005$ ). DPP significantly outperforms MMR on sequence-level overlap without any diversity hyperparameter.

**Fixed  $k = 3$ .** For apples-to-apples comparison with Lead-3:

| method   | ROUGE-1       | ROUGE-2       | ROUGE-L       |
|----------|---------------|---------------|---------------|
| Lead-3   | <b>0.1921</b> | <b>0.0295</b> | <b>0.1227</b> |
| MMR      | 0.1779        | 0.0293        | 0.1145        |
| DPP      | 0.1754        | 0.0265        | 0.1147        |
| Centroid | 0.1679        | 0.0285        | 0.1116        |
| Random   | 0.1682        | 0.0224        | 0.1066        |

Paired Wilcoxon of DPP vs. MMR at  $k = 3$ : all  $p > 0.25$  (statistically tied). DPP significantly beats random on ROUGE-L (+0.008,  $p = 0.024$ ). All content-aware methods lose to Lead-3 by  $\sim 0.8$  ROUGE-L points, consistent with the well-documented lead bias in news datasets [10].

**Interpretation.** DPP’s empirical advantage over MMR *appears at the variationally-selected size*, not at an arbitrary fixed  $k$ . When forced to  $k = 3$  both methods collapse to statistical parity; at the self-determined size they diverge in DPP’s favor. This indicates that the parameter-free size selection is itself part of the DPP form’s empirical contribution—not an incidental convenience—and is consistent with Theorem 3(iv): the log-det objective encodes a balance between  $|S|$  and pairwise similarity  $c$  whose optimum cannot be reconstructed by fixing  $|S|$  externally.

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